Kangmin Cho

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EDUCATION

Master of Science, Systems Engineering — Expected: May 2027
Cornell University, Ithaca, NY

Bachelor of Science/Master of Science, Applied Mathematics and Statistics — June 2021
Stony Brook University, Stony Brook, NY
Program: BS-MS Accelerated Program

Bachelor of Science, Technological Systems Management, — December 2020
Stony Brook University, Stony Brook, NY
Honor: Summa cum laude

RESEARCH EXPERIENCE

Graduate Researcher — Aug. 2025 – Present
Systems Engineering, Cornell University

- **Renewable Uncertainty Modeling for Green Hydrogen Systems**
  (Advisors: Prof. Charlle Sy, Prof. Michael Charles)
  - Designed and trained a stacked-LSTM conditional Generative Adversarial Network (cGAN) to synthesize 24-hour solar PV and wind power profiles.
  - Validated synthetic scenarios against Typical Meteorological Year, demonstrating the ability to capture interannual variability for a green hydrogen MILP optimization.
- **AI-Enabled Corporate Sustainability Assurance in the Energy Sector**
  (Advisors: Dr. Xinlai Liu, Prof. Oliver Gao)
  - Leading the development of a multimodal ESG assurance framework integrating Retrieval Augmented Generation (RAG) and Large Language Models (LLMs) to support sustainability auditing in the energy sector against international reporting standards.

Researcher — Jul. 2020 – Feb. 2023
Commodity Ecology Research Lab on the Circular Economy,
Stony Brook University, The State University of New York Korea *(Advisor: Prof. Mark Whitaker)*

- Conducted qualitative assessments of materials and technologies for computing hardware across sustainability impact categories, and developed industrial themes with actionable policy recommendations.

PUBLICATIONS

Manuscripts Under Review & In Preparation

[1] **Cho, K.**, Kim, M., Sun, Z., Charles, M., & Sy, C. A cGAN-MILP framework for renewable-powered hydrogen systems under renewable uncertainty. *Under second-round review at Digital Chemical Engineering.*

[2] **Cho, K.**, Zhang, Y., & Liu, X. AI-Enabled Sustainability Auditing: A RAG-Enhanced LLM Framework in the Energy Sector. *Manuscript in preparation.*

Peer-Reviewed Journal Articles

[3] **Cho, K.**, Jung, K., Lee, M., Lee, Y., Park, J., & Dreamson, N. (2022). Qualitative Approaches to Evaluating Social Media Influencers: A Case-Based Literature Review. *International Journal of Electronic Commerce Studies*, 13(2), 89–112. doi: `10.7903/ijecs.2025`

Book Chapters

[4] **Cho, K.** (Forthcoming). Computer Hardware, Capacitors, and Semiconductor Choices toward a Circular Economy. In M. D. Whitaker (Ed.), *Relinking Eden: The Best Material and Organizational Choices to Accelerate a Circular Economy.* Springer Nature SDG Series (SDG 12: Sustainable Production and Consumption). *(Contracted).*

SELECTED PROJECTS

Grid Decarbonization & Emerging Electricity Demand in NYS Fall 2026 – Present
SYSEN 5120 – Energy and Climate Scenario Analysis

- Developing long-term scenario models in LEAP to evaluate New York State (NYS) Climate Leadership and Community Protection Act (CLCPA) targets under rapid data-center load growth, building electrification, and EV adoption.

Spatiotemporal Environmental Audit of Shale Gas Infrastructure Spring 2026
PLSCI 5200 – Geographic Information Systems

- Examined 23-year forest ecosystem impacts (ΔNDVI, ΔLST) across 6,455 unconventional natural gas wells and 127 pipelines in three Marcellus Shale counties in Pennsylvania using Google Earth Engine (Landsat 5/9, NLCD, SRTM), ArcGIS Pro, and QGIS.

TEACHING EXPERIENCE

Course Assistant & Grader Sep. 2025 – Present
Cornell University & Weill Cornell Medicine

- Facilitated student Q&A, graded assignments, and evaluated team projects across three graduate courses: *Foundations of Systems Engineering* (Fall 2026), *Healthcare Data Management* (Spring 2026), *Health Data Management and Analytics* (Fall 2025).

Adjunct Instructor Jan. 2025 – Present
Camp Humphreys, University of Maryland Global Campus

- Sessional instructor for STAT 200: Introduction to Statistics at USAG-Humphreys.

Adjunct Instructor Fall 2023 – Spring 2024
Department of Applied Mathematics and Statistics,
Stony Brook University, The State University of New York Korea

- Taught AMS 104: Spreadsheet Technology and Applications, guiding students in Excel-based data analysis and management.
- Mentored students in data collection, statistical methods, and visualization, improving their applied analytics skills.

Instructor Summer 2023, Summer 2024
Fashion Institute of Technology, The State University of New York Korea

- Taught college algebra, introductory statistics, and geometry to incoming students in the Access to Excellence (AE) Program.
- Developed course curricula, instructional materials, quizzes, and assignments in alignment with departmental standards.

HONORS AND AWARDS

OTEFE Foundation Graduate Grant, OTEFE Foundation, Inc., 2025
Teaching Assistantship, Stony Brook University, 2020-2021
NH Bank Scholarship, National Agricultural Cooperative Federation, Republic of Korea, 2019
SUNY Academic Prestige Scholarship, The State University of New York Korea, 2018-2019
SUNY Academic Excellence Scholarship, The State University of New York Korea, 2017
Academic Achievement Award, Stony Brook University, 2017

UNIVERSITY SERVICE & LEADERSHIP

Treasurer Dec. 2025 – Present
Systems Engineering Graduate Student Association, Cornell University

- Manage budgets, oversee financial records of the association, and assist monthly social for graduate students.

Student Academic Services Assistant Sep. 2025 – Present
Hasbrouck Community Center, Cornell University

- Supporting daily operations and providing customer service to maintain a welcoming service center environment.

SKILLS

Technical Skills:
Programming: Python (PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib), R, SAS, SQL
Energy Systems Modeling: LEAP (Low Emissions Analysis Platform)
Geospatial Analysis: ArcGIS Pro, QGIS, Google Earth Engine